Time-Series Analysis of COVID-19 Data and Unsupervised Digit Clustering Evaluation

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1. ABSTRACT

This study presents a data engineering workflow that covers two main areas: large-scale epidemiological time-series analysis and unsupervised pattern recognition using pixel datasets. The first phase of the project analyzes global health trends using the official World Health Organization (WHO) COVID-19 daily dataset. Using vectorized data transformations in Python and Pandas, over 491,000 cases were processed. The dataset was trimmed to a specific date range between March 2020 and August 2023. This allowed for granular temporal filtering, regional group aggregations, quarterly case summarization, and pivot matrix generation to track virus transmission across different global zones.

The second phase evaluates unsupervised machine learning techniques on high-dimensional data. The MNIST handwritten digit database was loaded, separating 1,797 image arrays into discrete numerical features. A K-Means clustering algorithm was applied to group the unlabeled data into 10 distinct target clusters. To handle the arbitrary index labels assigned by K-Means, an optimal label-mapping technique using cluster modes was deployed to align predictions with true digits.

The performance of the model was evaluated using macro-averaged F1 metrics, achieving a final validation score of 0.8628. The results demonstrate that combining structured data preprocessing with unsupervised statistical clustering provides a reliable pipeline for handling both tabular time-series data and complex image patterns.

2. INTRODUCTION

2.1 Relevance of the Project

The modern data engineering field requires the ability to extract actionable insights from both unstructured data patterns and massive public health time-series logs. The first part of this project addresses the computational challenges of monitoring global health crises. Processing massive, multi-region epidemiological records requires efficient, vectorized data pipelines. By building automated preprocessing workflows, data scientists can track transmission vectors, spot regional variations, and provide verified information to healthcare teams.

The second part of the project addresses image recognition, where hand-drawn patterns must be sorted without prior labeling. This is achieved using unsupervised clustering algorithms. Testing these systems on historical baselines like the MNIST dataset helps developers understand model limits, mapping issues, and boundary separation challenges. These skills are essential for deploying automated diagnostic software and document routing networks.

2.2 Technology and Libraries Involved

- **Python Engine:** The core environment used to run data pipelines and statistical equations.
- **Pandas API:** Used for high-speed tabular operations, datetime index mapping, quarterly period conversions, and multi-axis pivot table grouping.
- **Scikit-Learn Framework:** The primary machine learning framework used to load datasets, initialize unsupervised cluster models, and compute macro F1 scoring reports.
- **SciPy Stats Engine:** Used to execute mode evaluations for cluster label mapping.

- **Google Colab Environment:** The cloud-based development environment used to run the code cells and store data arrays securely.

2.3 Background Material Survey and Data Sources

This research utilizes two separate data repositories:

1. **WHO COVID-19 Global Daily Data:** A massive tabular dataset tracked by the World Health Organization. It contains historical records for individual nations, regional codes, daily new counts, and cumulative summaries for both cases and fatalities.
2. **MNIST Digits Dataset:** A benchmark dataset consisting of low-resolution pixel grids representing handwritten digits from 0 to 9. Each record contains 64 independent features representing a flattened 8x8 matrix, serving as a core testing framework for computer vision models.

2.4 Technical Training Received

During the initial weeks of the summer program at IDEAS Foundation, ISI Kolkata, key training areas were covered:

- Object serialization, secure remote dataset fetching, and file streaming methods inside Google Colab notebook instances.
- Rectifying `SettingWithCopyWarning` issues by handling data slice references correctly during feature type assignments.
- Time-series processing methods, including converting string values into proper `datetime64[ns]` datatypes and handling quarterly and monthly periods.
- Unsupervised learning algorithms, focusing on centroid optimization dynamics, cluster inertia limits, and mode-based label mapping strategies.

3. PROJECT OBJECTIVE

The primary engineering objectives of this project include:

- **Time-Series Pipeline Automation:** Building a clean, end-to-end framework to load, clean, convert, and subset raw public health datasets.
- **Temporal Trend Analysis:** Extracting regional summaries, finding dates with peak infections, and analyzing quarterly shifts.
- **Multi-Dimensional Matrix Generation:** Building cross-tabulated pivot tables to map geographical region rows against discrete monthly columns.
- **Unsupervised Model Implementation:** Training a 10-cluster K-Means model on high-dimensional pixel matrices.
- **Clustering Mapping and Evaluation:** Designing a mode-matching mapping system to calculate accurate macro F1 validation scores.

4. METHODOLOGY

- ## 4.1 Data Pipeline Engineering Flowchart
- The structural flow of data processing operations followed a clear pipeline architecture:
- `[Phase 1: Time Series Processing]`
- `Load WHO Data ➔ Convert String to Datetime ➔ Trim Range (2020-2023) ➔ Regional Pivot & Aggregation`

- `[Phase 2: Unsupervised Learning]`

- Load MNIST Data ➔ Extract 64-Pixel Features ➔ Fit 10-Cluster K-Means ➔ Mode Label Mapping ➔ Macro F1 Scoring

- ## 4.2 Data Collection and Preprocessing Steps

- The primary epidemiological dataset was ingested directly from a verified cloud directory using native Python file utilities. The raw dataframe structure includes specific data columns, including `Date_reported`, `Country_code`, `Country`, `WHO_region`, `New_cases`, `Cumulative_cases`, `New_deaths`, and `Cumulative_deaths`.
- To optimize processing, a subset was created using five core columns: `Date_reported`, `Country`, `WHO_region`, `New_cases`, and `New_deaths`. The `Date_reported` column, originally parsed as a generic string object, was converted using `pd.to_datetime()` into a high-precision `datetime64[ns]` timestamp. This conversion enables temporal slicing and avoids performance issues during data analysis.
- ## 4.3 Temporal Trimming and Filtering Strategy
- A strict time filter was applied to remove incomplete boundary values from the data. The dataframe was filtered to include only rows between March 1, 2020, and August 31, 2023, inclusive. This filtering resulted in a clean dataset of exactly 306,960 operational records ready for statistical analysis.
- ## 4.4 Unsupervised Clustering and Mapping Math
- For the unsupervised learning component, the MNIST data array was loaded into a feature space with dimensions of 1797 rows by 64 attributes. Because the data labels were removed during training, the K-Means algorithm was initialized to find 10 clusters using a fixed seed of `random_state=42` to ensure consistent cluster results.
- Since K-Means assigns cluster labels arbitrarily, a mapping process was applied to evaluate the model's accuracy. The pipeline iterates through each cluster, applies a Boolean mask to locate the corresponding samples, and calculates the true statistical mode using `scipy.stats.mode`. This mode value is then assigned to the cluster, mapping the arbitrary cluster labels back to their true handwritten digit meanings.

DATA ANALYSIS AND RESULTS

5.1 Analysis of COVID-19 Time-Series Records

5.1.1 Initial Data Ingestion and Validation

The initial data load generated a large, multi-column matrix with 491,040 rows. Early rows reveal data variations from the start of the recording period in January 2020:

Record Row	Date Reported	Country Name	Region Code	New Cases	New Deaths
0	2020-01-04	Anguilla	AMR	*NaN*	*NaN*
1	2020-01-04	Azerbaijan	EUR	*NaN*	*NaN*
2	2020-01-04	Bangladesh	SEAR	0.0	0.0
3	2020-01-04	Barbados	AMR	*NaN*	*NaN*

4	2020-01-04	Brazil	AMR	*NaN*	*NaN*

5.1.2 Analysis of Top Impacted Nations

Grouping by country and tracking maximum cumulative volumes identified the 5 countries with the highest total reported cases:

1. **United States of America:** 103,436,829 total cases
2. **China:** 99,381,761 total cases
3. **India:** 45,055,954 total cases
4. **France:** 39,042,805 total cases
5. **Germany:** 38,437,875 total cases

5.1.3 Regional Analysis and Quarterly Aggregations

Summing case counts across different regions highlighted the areas most affected during the tracking period:

- **Europe (EUR):** 275,983,896 total cases
- **Western Pacific (WPR):** 213,827,362 total cases
- **Americas (AMR):** 192,901,815 total cases
- **South-East Asia (SEAR):** 54,388,896 total cases
- **Eastern Mediterranean (EMR):** 23,388,717 total cases
- **Africa (AFR):** 9,549,965 total cases
- **Other Locations:** 65 total cases

Converting timelines into quarterly periods shows how case numbers rose during the initial phase of the pandemic:

Quarterly Period	Total Quarterly New Cases	Total Quarterly New Deaths
2020Q1	697,567.0	40,666.0
2020Q2	9,377,665.0	511,574.0
2020Q3	23,887,893.0	546,584.0
2020Q4	48,287,242.0	834,992.0
2021Q1	45,249,877.0	1,026,874.0

5.1.4 Global Peak Identification

Using index optimization functions on the aggregated daily global cases dataframe identified the single day with the highest number of new infections worldwide:

- **Peak Date:** January 30, 2022
- **Global New Cases Count on Peak Day:** 8,401,963.0 cases

5.1.5 Monthly Cross-Tabulation Pivot Output

The table below tracks regional trends across different stages of the pandemic, showing total monthly cases by region:

WHO Region	2020-03 Cases	2020-06 Cases	2021-12 Cases	2022-12 Cases	2023-08 Cases
AFR	4,042.0	200,688.0	*High Peak*	38,914.0	2,902.0
AMR	197,767.0	2,418,310.0	*High Peak*	4,230,321.0	13,176.0
EMR	49,425.0	477,341.0	*Mid Peak*	27,172.0	3,698.0
EUR	419,584.0	535,296.0	*High Peak*	4,378,974.0	172,170.0
SEAR	3,058.0	496,321.0	*Mid Peak*	24,171.0	3,845.0
WPR	23,678.0	63,812.0	*Low Bound*	82,283,226.0	1,240,640.0

5.2 Analysis of Image Clustering Results

The K-Means clustering algorithm was applied to the MNIST dataset, processing 1,797 image samples across 64 feature dimensions. The model grouped the data points into 10 target categories.

The mapped cluster array was evaluated against the true digit classes using a macro-averaged F1 validation script. The model achieved a final macro F1 score of **0.8628**. This indicates strong clustering performance, showing that the model could reliably distinguish complex handwritten structures without prior exposure to labels.

6. CONCLUSION

6.1 Core Findings and Project Summary

This project successfully demonstrates data pipelines for both public health time-series analysis and unsupervised pattern clustering. The time-series pipeline handled a large, multi-region dataset, enabling efficient sorting, regional grouping, and cross-tabulation filtering. The analysis highlighted key trends, including a global peak on January 30, 2022, with over 8.4 million new cases in a single day.

In the unsupervised learning phase, the K-Means model grouped the high-dimensional MNIST dataset with a macro F1 score of 0.8628. This highlights the effectiveness of mode-based label mapping for evaluating unsupervised models.

6.2 Recommendations for Future Work

- Integrate automated rolling-average lines into the time-series pipeline to smooth out daily reporting noise.
- Test more advanced clustering models, such as Hierarchical Density-Based Spatial Clustering (HDBSCAN), to improve handling of overlapping digits.
- Optimize processing speeds for large datasets by migrating the code to distributed computing platforms like PySpark.

7. APPENDICES

Appendix 1: References

- World Health Organization (WHO) COVID-19 Public Repository Tracking Data (2026).
- MacQueen, J. (1967). "Some methods for classification and analysis of multivariate observations." *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*.
- Pandas and Scikit-Learn API documentation for data transformation and clustering modules.

Appendix 2: Digital Asset Repositories

- **GitHub Repository Link (Public):** https://github.com/aninditamazumdar79-cmyk/Time-Series-Analysis-of-COVID-19-Data.git
- **Notebook GitHub Link (Direct File):** https://github.com/aninditamazumdar79-cmyk/Time-Series-Analysis-of-COVID-19-

Data/blob/main/06 Time Series Analysis of COVID 19 Data Summer 2026.ipynb